

Name: _____

Date: _____

DIRECTIONS

This Independent Practice is for Lesson 4.11.05.

Show your work in the box. Write your final answer on the line.

Read each problem carefully. Use a model or picture if it helps.

If you get stuck, re-read the question and look at any example you already worked through in the lesson.

1 L-shape perimeter

An L-shape: outer $7\text{ cm} \times 5\text{ cm}$ with a $3\text{ cm} \times 2\text{ cm}$ notch cut from top-right.

(a) Find the missing top horizontal piece ($7 - 3 = \underline{\quad}$).

(b) Find the missing right vertical piece ($5 - 2 = \underline{\quad}$).

(c) Add all six sides to find the perimeter.

LABEL ALL SIDES; SUM

(a) ____ (b) ____ (c) ____ cm

2 Cool fact — bounding rectangle

For the L-shape in Problem 1:

- (a) What is the perimeter of the OUTER BOUNDING RECTANGLE (7×5)?
- (b) Compare to your answer in Problem 1c. What do you notice?
- (c) Explain WHY the L-shape's perimeter equals the bounding rectangle's perimeter.

COMPARE; EXPLAIN

(a) ____ cm (b) ____ (c) ____

3 T-shape — eight sides

A T-shape: top bar $12 \text{ cm} \times 3 \text{ cm}$, stem $4 \text{ cm} \times 5 \text{ cm}$ (centered under the bar).

- (a) Find the two missing side pieces of the top bar (under the bar, on each side of the stem).
- (b) Add all 8 sides to find the perimeter.

FIND MISSING; SUM

(a) ____ (b) ____ cm

4 Stairs

A staircase composite figure has 3 equal steps. Each step is 2 cm wide and 2 cm tall. The total outer width is 6 cm and outer height is 6 cm.

- (a) How many sides does this figure have?
- (b) Find the perimeter by adding all sides.
- (c) What is the perimeter of the bounding 6×6 square? Compare.

COUNT SIDES; SUM; COMPARE

(a) ____ (b) ____ cm (c) ____

5 Word problem — Asher's L-shaped rug

Asher has an L-shaped rug. Outer dimensions are 8 ft by 6 ft. A 3 ft by 2 ft notch is cut from one corner.

- (a) Find the perimeter of the rug (add all 6 sides).
- (b) Find the area of the rug.
- (c) Asher wants to stitch a ribbon along the entire edge of the rug. How many feet of ribbon does he need?

PERIMETER; AREA; RIBBON

(a) ____ ft (b) ____ sq ft (c) ____ ft

6 Find and fix

A student found the perimeter of an L-shape by computing the OUTER rectangle's area (length $\times$ width).

- (a) Why is this wrong (two ways)?
(b) What method should they have used?

ANALYZE THE ERROR

(a) ____ (b) ____